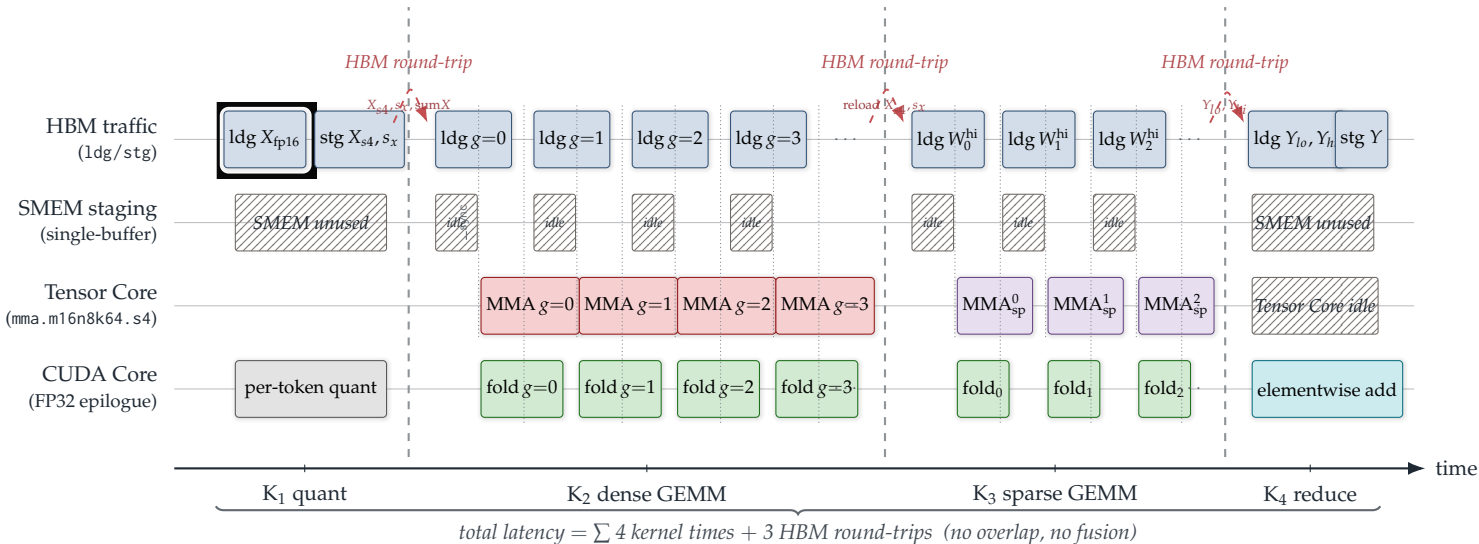


K₁: activation_quantK₂: dense_gemmK₃: sparse_gemmK₄: reduce_sum

K₂ stall pattern: each quant-group does ldg → ...sync → MMA → ...sync – no cp.async, no double-buffer

Legend:



activation quant

FP32 epilogue fold



HBM ldg / stg

reduce_sum



INT4 Tensor Core (dense)

idle / ...sync threads stall



INT4 Tensor Core (sparse BSR)

--▶ HBM round-trip between kernels